

1 Introduction

TBD

2 Equations of Motion

2.1 Analytical Derivation

Consider a ball of radius R and mass M rolling without slipping on a plane

$$H := \{(x, y, z) \in \mathbb{R}^3 \mid F := z - \alpha y = 0\} \subseteq \mathbb{R}^3 \quad (1)$$

with unit normal field $\mathbf{n} := \frac{\text{grad } F}{\|\text{grad } F\|}$. Additionally, assume that the sphere carries a dipole moment $\boldsymbol{\mu}_0$ interacting with a homogeneous, constant magnetic field $\mathbf{B}$. The configuration manifold is

$$Q := H \times \text{SO}(3) \subseteq \mathbb{R}^3 \times \text{SO}(3),$$

and the velocity phase space is therefore given by

$$TQ = TH \times T\text{SO}(3) \cong TH \times \mathfrak{so}(3) \cong \mathbb{R}^2 \times \mathbb{R}^3 \subseteq \mathbb{R}^6.$$

We choose the trivial global chart on $\mathbb{R}^3$ and some local smooth chart $\phi : U \subseteq \text{SO}(3) \rightarrow \mathbb{R}^3$ on $\text{SO}(3)$.¹ A motion is a function

$$(\mathbf{r}(t), P(t)) : \mathbb{R} \supseteq I \rightarrow \mathbb{R}^3 \times \text{SO}(3).$$

In local coordinates, define $\boldsymbol{\theta} := \phi \circ P : I \rightarrow \mathbb{R}^3$. Our goal is to find equations of motion and their solutions

$$(\mathbf{r}(t), \boldsymbol{\theta}(t)) = (x(t), y(t), z(t), \theta_x(t), \theta_y(t), \theta_z(t)) : I \rightarrow \mathbb{R}^6$$

in local coordinates. Using the canonical Euclidean basis $\mathcal{B} := (\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, we define a non-moving (inertial) coordinate frame by choosing an arbitrary origin. Furthermore, we consider a rotating coordinate frame defined by $\hat{\mathcal{B}}(t) := P(t)\mathcal{B}$. The angular velocity in the inertial frame $\boldsymbol{\Omega}(t) = (\Omega_x, \Omega_y, \Omega_z)^\top$ is defined in terms of the cross product matrix

$$\Omega^\times = \dot{P}P^\top = \begin{pmatrix} 0 & -\Omega_z & \Omega_y \\ \Omega_z & 0 & -\Omega_x \\ -\Omega_y & \Omega_x & 0 \end{pmatrix}.$$

¹Constraining the motion to H reduces the system by one dimension, but we chose not to eliminate z since we need the solution in the full frame for numerical analysis.

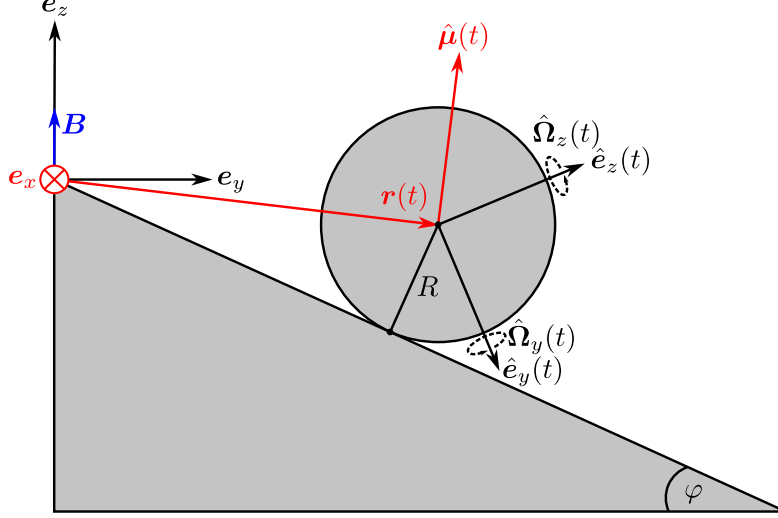


Figure 1: Sketch of the system in the yz -plane. The hat vectors are vectors of the basis $\hat{\mathcal{B}}$ while vectors without hat refer to $\mathcal{B}$. Red vectors are projected and may have non-zero e_x -components. The hyperplane has an angle of inclination φ defined by $\alpha = \tan \varphi$. The outer magnetic field is kept fixed parallel to e_z .

The kinetic energy form is given by

$$T(\mathbf{r}, \mathbf{\Omega}) = \frac{M}{2} \|\dot{\mathbf{r}}\|^2 + \frac{I_0}{2} \|\mathbf{\Omega}\|^2 \quad (2)$$

and the potential by

$$U(z, \mathbf{\Omega}) = -Mgz + \langle \hat{\boldsymbol{\mu}}(t), \mathbf{B} \rangle, \quad (3)$$

where $\hat{\boldsymbol{\mu}}(t) := P(t)\boldsymbol{\mu}_0$, I_0 is the inertia scalar of an ideal ball, and g is the gravitational acceleration. We obtain a *Lagrangian* $L(z, \mathbf{\Omega}, \dot{\mathbf{r}}) := T - U$. Assuming that there is an external force $\mathbf{F}_e = \mathbf{F}_e(\mathbf{r})$ such as friction, the corresponding *Lagrange-d'Alembert principle*[3][5] reads

$$\delta \int_0^t L(z(s), \mathbf{\Omega}(s), \dot{\mathbf{r}}(s)) ds + \int_0^t \langle \mathbf{F}_e(\mathbf{r}(s)), \delta \mathbf{r}(s) \rangle ds = 0. \quad (4)$$

The torque can be written as

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{em}} + \boldsymbol{\tau}_e$$

where $\boldsymbol{\tau}_{\text{em}} := \hat{\boldsymbol{\mu}}(t) \times \mathbf{B}$ is the magnetic torque, and $\boldsymbol{\tau}_e := -R\mathbf{n} \times \mathbf{F}_e$ is the torque induced by external forces. The phase space of the linear motion admits an orthogonal decomposition $\mathbb{R}^3 \cong H \oplus H^\perp$. Thus we can decompose arbitrary $\mathbf{v} \in \mathbb{R}^3$ as $\mathbf{v} = \mathbf{v}_\parallel + v_\perp \mathbf{n}$. With the rolling constraint

$$\dot{\mathbf{r}} = R\boldsymbol{\Omega} \times \mathbf{n}, \quad (5)$$

integration by parts, and non-degeneracy of the scalar product, one obtains tangential

$$\frac{d\boldsymbol{\Omega}_\parallel}{dt} = \frac{1}{I_0 + MR^2} \boldsymbol{\tau}_\parallel - \frac{RMg}{I_0 + MR^2} (\mathbf{n} \times \mathbf{e}_z) \quad (6)$$

and normal

$$\frac{d\Omega_\perp}{dt} = \frac{\tau_\perp}{I_0} \quad (7)$$

equations for the angular acceleration from Equation 4. Taking the derivative of Equation 5, this fully determines the linear motion. Assuming Coulomb friction[2][4] and aerodynamical drag by the drag equation[Stokes2009], the external force becomes

$$\mathbf{F}_e = -\nu_r Mg \langle \mathbf{n}, \mathbf{e}_z \rangle \text{sgn}(\dot{\mathbf{r}}) + \frac{\rho CA}{2} \|\dot{\mathbf{r}}\|^2, \quad (8)$$

where ν_r is the rolling resistance coefficient, sgn is the sign function, ρ is the fluid density of air, C is the drag coefficient, and A is the cross-sectional area of the sphere. For numerical reasons, we reparametrize $P(t) = P((q_0(t), \mathbf{q}(t)))$, where

$$(q_0(t), \mathbf{q}(t)) : I \rightarrow \mathbb{S}^3$$

is a time-dependent unit quaternion. Subsuming, the full equations of motion are:

$$\begin{aligned} \frac{d^2 \mathbf{r}}{dt^2} &= \frac{5}{7M} \boldsymbol{\tau} \times \mathbf{n} - \frac{5g}{7} (\mathbf{e}_z)_\parallel, \\ \frac{d\boldsymbol{\Omega}_\parallel}{dt} &= \frac{5}{7MR^2} \boldsymbol{\tau}_\parallel - \frac{5g}{7R} \mathbf{n} \times \mathbf{e}_z, \\ \frac{d\Omega_\perp}{dt} &= \frac{5}{7MR^2} \tau_\perp, \\ \frac{d(q_0, \mathbf{q})}{dt} &= \frac{1}{2} (0, \boldsymbol{\Omega}) \cdot (q_0, \mathbf{q}). \end{aligned} \quad (9)$$

2.2 Numerical Solution

The equations of motion were solved with our program `magsphere`[7]. We implemented a Runge-Kutta-Munthe-Kaas-4 method, combining a quaternion-adapted integrator with the Runge-Kutta 4 algorithm.[1][6] To avoid numerical instability at the discontinuities in Equation 8, we approximate

$$\text{sgn}(\dot{\mathbf{r}}) \approx \tanh(k\|\dot{\mathbf{r}}\|),$$

where k is a dimensionless constant describing the steepness of $\tanh$ around the origin.

3 Results

3.1 Influence of the magnetic moment

We investigated the influence of the initial magnetic moment vector $\boldsymbol{\mu}_0$ on the trajectories for an incline angle of $\varphi = 10^\circ$. Firstly focussing on the influence of the deviation of $\boldsymbol{\mu}_0$ from the plane spanned by $\mathbf{B}$ and the downhill direction, we parametrized

$$\boldsymbol{\mu}_0 = \mu_0(\sin \theta, \cos \theta, 0)^\top$$

with $\mu_0 := \|\boldsymbol{\mu}_0\|$ and the polar angle θ defined as depicted in Figure ??.

4 References

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